Domácí úloha 1

Zadání

Pro $n \in \mathbb{N}$ určete:

$$\begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}^n$$

Řešení

$$A = \begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}$$

$$p_A(\lambda) = \begin{vmatrix} -1-\lambda & 6 \\ -1 & 4-\lambda \end{vmatrix} = \lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2)$$

$$\lambda_1 = 1$$

$$\lambda_2 = 2$$

$$V_1 = \mathrm{Ker} \begin{pmatrix} -2 & 6 \\ -1 & 3 \end{pmatrix} = \left\langle (3,1)^T \right\rangle$$

$$V_2 = \mathrm{Ker} \begin{pmatrix} -3 & 6 \\ -1 & 2 \end{pmatrix} = \left\langle (2,1)^T \right\rangle$$

$$\left(\begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 3 & 3 & 0 & 3 \end{array} \right) \sim \left(\begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 0 & 1 & -1 & 3 \end{array} \right)$$

$$\sim \left(\begin{array}{cc|cc} 3 & 0 & 3 & -6 \\ 0 & 1 & -1 & 3 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & -1 & 3 \end{array} \right)$$

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$$

Výsledek

$$\begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}^n = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$$

Domácí úloha 2

Zadání

Diagonalizujte matici:

$$A = \begin{pmatrix} 2 & 3 & 3 \\ 2 & 1 & 2 \\ -4 & -4 & -5 \end{pmatrix}$$

Řešení

$$p_A(\lambda) = \begin{vmatrix} 2-\lambda & 3 & 3 \\ 2 & 1-\lambda & 2 \\ -4 & -4 & -5-\lambda \end{vmatrix} = (-\lambda)^3 + \mathrm{Tr}A(-\lambda)^2 + \mathrm{Tr}(\mathrm{adj}A)(-\lambda) + \det A$$

$$\mathrm{adj}A = \begin{pmatrix} 3 & * & * \\ * & 2 & * \\ * & * & -4 \end{pmatrix}$$

$$\det A = -10 - 24 - 24 + 12 + 16 + 30 = 0$$

$$p_A(\lambda) = -\lambda^3 - 2\lambda^2 - 1\lambda$$

$$\lambda^3 + 2\lambda^2 + \lambda = 0$$
$$\lambda(\lambda+1)^2 = 0$$

$$\lambda_1 = 0$$

$$\lambda_2 = \lambda_3 = -1$$

$$V_0 = \operatorname{Ker} \begin{pmatrix} 2 & 3 & 3 \\ 2 & 1 & 2 \\ -4 & -4 & -5 \end{pmatrix} = \operatorname{Ker} \begin{pmatrix} 0 & 2 & 1 \\ 2 & 1 & 2 \\ 0 & -2 & -1 \end{pmatrix}$$

$$= \operatorname{Ker} \begin{pmatrix} 2 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \left\langle (3,2,-4)^T \right\rangle$$

$$V_{-1} = \operatorname{Ker} \begin{pmatrix} 3 & 3 & 3 \\ 2 & 2 & 2 \\ -4 & -4 & -4 \end{pmatrix} \begin{array}{l} \mid \cdot \frac{1}{3} \\ \mid \cdot \frac{1}{2} \\ \mid \cdot -\frac{1}{4} \end{array}$$

$$= \operatorname{Ker} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \left\langle (1,0,-1)^T, (1,-1,0)^T \right\rangle$$

$$\left(\begin{array}{ccc|ccc} 3 & 1 & 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ -4 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & -1 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 0 & 2 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & -1 & 0 \\ 0 & -2 & -5 & -2 & 3 & 0 \\ 0 & -1 & -2 & 0 & 2 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & -1 & 0 \\ 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & -2 & -1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & -3 & -3 & -4 \\ 0 & 1 & 0 & -4 & -4 & -5 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & -4 & -4 & -5 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{array}\right)$$

Výsledek

$$\begin{pmatrix} 2 & 3 & 3 \\ 2 & 1 & 2 \\ -4 & -4 & -5 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 0 & -1 \\ -4 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -4 & -4 & -5 \\ 2 & 1 & 2 \end{pmatrix}$$

Domácí úloha 3

Zadání

Uvažujme matici A. Najděte všechny matice X takové, že AX je násobkem matice X:

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Řešení

Nechť:

$$X = x_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$[X]^B = (x_1, x_2, x_3, x_4)^T$$

a dále nechť:

$$f(X) = AX,$$

pak:

$$[f]_B^B = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Pak hledáme taková X, aby platilo $f(X) = \lambda X$, tedy:

$$[X]^B \in \text{Ker } [(AX - \lambda E)]^B$$

$$p_f(\lambda) = \begin{vmatrix} 1-\lambda & 0 & -1 & 0 \\ 0 & 1-\lambda & 0 & -1 \\ -1 & 0 & 1-\lambda & 0 \\ 0 & -1 & 0 & 1-\lambda \end{vmatrix} =$$

$$= (1-\lambda) \begin{vmatrix} 1-\lambda & 0 & -1 \\ 0 & 1-\lambda & 0 \\ -1 & 0 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & -1 & 0 \\ 1-\lambda & 0 & -1 \\ -1 & 0 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)^2 \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix}$$

$$= (\lambda^2 - 2\lambda)(\lambda^2 - 2\lambda) = \lambda^2(\lambda - 2)^2$$

$$\lambda_1 = 0$$
$$\lambda_2 = 2$$

$$V_0 = \text{Ker} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} = \left\langle (1,0,1,0)^T, (0,1,0,1)^T \right\rangle$$

$$V_2 = \text{Ker} \begin{pmatrix} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix} = \left\langle (1,0,-1,0)^T, (0,1,0,-1)^T \right\rangle$$

Výsledek

$$X_0 \in \left\langle \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \qquad AX_0 = 0$$

$$X_2 \in \left\langle \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} \right\rangle \qquad AX_2 = 2X_2$$

Domácí úloha 4

Zadání

Ukažte, že charakteristický polynom matice A je až na násobek roven $\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$, a tedy že ke každému polynomu $p(\lambda)$ existuje matice, jejímž charakteristickým polynomem $p(\lambda)$ je:

$$A = \begin{pmatrix} -a_{n-1} & 1 & 0 & 0 & \dots & 0 \\ -a_{n-2} & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ -a_2 & 0 & \dots & 0 & 1 & 0 \\ -a_1 & 0 & \dots & 0 & 0 & 1 \\ -a_0 & 0 & \dots & 0 & 0 & 0 \end{pmatrix}$$

Řešení

$$p_A(\lambda) = \begin{vmatrix} -a_{n-1} - \lambda & 1 & 0 & 0 & \dots & 0 \\ -a_{n-2} & -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ -a_2 & 0 & \dots & -\lambda & 1 & 0 \\ -a_1 & 0 & \dots & 0 & -\lambda & 1 \\ -a_0 & 0 & \dots & 0 & 0 & -\lambda \end{vmatrix}$$

$$= (-a_{n-1}-\lambda)\begin{vmatrix} -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \dots & -\lambda & 1 & 0 \\ 0 & \dots & 0 & -\lambda & 1 \\ 0 & \dots & 0 & 0 & -\lambda \end{vmatrix} + (-1)\begin{vmatrix} -a_{n-2} & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ -a_2 & \dots & -\lambda & 1 & 0 \\ -a_1 & \dots & 0 & -\lambda & 1 \\ -a_0 & \dots & 0 & 0 & -\lambda \end{vmatrix}$$

$$\begin{vmatrix} -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \dots & -\lambda & 1 & 0 \\ 0 & \dots & 0 & -\lambda & 1 \\ 0 & \dots & 0 & 0 & -\lambda \end{vmatrix} = \begin{vmatrix} -\lambda & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \dots & -\lambda & 0 & 0 \\ 0 & \dots & 0 & -\lambda & 0 \\ 0 & \dots & 0 & 0 & -\lambda \end{vmatrix} = (-\lambda)^{n-1}$$

$$\begin{aligned} p_A(\lambda) &= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + (-1)(-a_{n-2})(-\lambda)^{n-2} + \\ &+ (-1)^2 \begin{vmatrix} -a_{n-3} & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ -a_2 & \dots & -\lambda & 1 & 0 \\ -a_1 & \dots & 0 & -\lambda & 1 \\ -a_0 & \dots & 0 & 0 & -\lambda \end{vmatrix} \\ &= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + (-1)(-a_{n-2})(-\lambda)^{n-2} + \dots + \\ &+ (-1)^{(n-3)}(-a_2)(-\lambda)^2 + (-1)^{(n-2)} \begin{vmatrix} -a_1 & 1 \\ -a_0 & -\lambda \end{vmatrix} \\ &= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + (-1)(-a_{n-2})(-\lambda)^{n-2} + \dots + \\ &+ (-1)^{(n-3)}(-a_2)(-\lambda)^2 + (-1)^{n-2}(-a_1)(-\lambda) + (-1)^{(n-1)}(-a_0) \\ &= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + \dots + (-1)^n a_1 \lambda + (-1)^n a_0 \end{aligned}$$

Výsledek

$$\forall n = 2k,\ k \in \mathbb{N} \qquad \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = p_A(\lambda)$$
$$\forall n = 2k-1,\ k \in \mathbb{N} \qquad \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = -p_A(\lambda)$$

Domácí úloha 5

Zadání

Reálná 5×5 matice A má spektrum $\sigma(A) = \{1, 2, 3, 4, 5\}$. Určete spektrum matice $\mathrm{adj}(A)$.

Řešení

Pakliže má matice A spektrum $\sigma(A) = \{1, 2, 3, 4, 5\}$, bude její diagonální matice D vypadat následovně:

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}$$

Determinant A pak musí být:

$$\det(A) = 120$$

Pak platí:

$$\begin{aligned} A &= RDR^{-1} \\ E &= A^{-1}RDR^{-1} \\ RD^{-1} &= A^{-1}R \\ D^{-1} &= R^{-1}A^{-1}R \\ D^{-1} &= R^{-}(\det(A))^{-1}\mathrm{adj}(A)R \\ \det(A)D^{-1} &= R^{-1}\mathrm{adj}(A)R \end{aligned}$$

Přičemž matice $R^{-1}\mathrm{adj}(A)R$ představuje diagonální matici $D_{\mathrm{adj}(A)}$ k matici $\mathrm{adj}(A)$, na jejíž diagonále se nachází její vlastní čísla, a tedy prvky jejího spektra.

$$D_{\mathrm{adj}(A)} = \det(A)D^{-1} = 120 \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 120 & 0 & 0 & 0 & 0 \\ 0 & 60 & 0 & 0 & 0 \\ 0 & 0 & 40 & 0 & 0 \\ 0 & 0 & 0 & 30 & 0 \\ 0 & 0 & 0 & 0 & 24 \end{pmatrix}$$

Výsledek

$$\sigma(\mathrm{adj}(A)) = \{120, 60, 40, 30, 24\}$$